

Polarization of Gravitational Waves From Turbulence: Tension Between Simulation and Analytic Approaches

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I. KRAICHNAN–K41 MODEL (GOGOBERIDZE 2007)

We adopt the spatial Fourier pair

$$R_{ij}(\mathbf{r}, \tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k}, \tau), \quad F_{ij}(\mathbf{k}, \tau) = \int d^3r e^{-i\mathbf{k}\cdot\mathbf{r}} R_{ij}(\mathbf{r}, \tau), \quad (1)$$

and the temporal Fourier pair

$$f(\eta_k, \tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k, \omega), \quad \tilde{f}(\eta_k, \omega) = \int d\tau e^{i\omega\tau} f(\eta_k, \tau). \quad (2)$$

Starting with all the definitions from Eqs. 23–26 of [1],

$$\begin{aligned} R_{ij}(\mathbf{r}, \tau) &= \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k}, \tau) \\ &= \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} A(k) P_{ij}(\hat{k}) f(\eta_k, \tau) \\ &= \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} A(k) P_{ij}(\hat{k}) \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k, \omega). \end{aligned} \quad (3)$$

Here $A(k) = C_k \epsilon^{2/3} k^{-11/3} / 4\pi$, $P_{ij}(\hat{k}) = \left(\delta_{ij} - \frac{k_i k_j}{k^2} \right)$, $f(\eta_k, \tau) = \exp(-\frac{\pi}{4} \eta_k^2 \tau^2)$, and $\eta_k = \frac{1}{\sqrt{2\pi}} \epsilon^{1/3} k^{2/3}$. Equation 28 of [1] gives

$$R_{ijij}(\mathbf{x}', \mathbf{x}' + \mathbf{r}, \tau) = R_{ii}(\mathbf{r}, \tau) R_{jj}(\mathbf{r}, \tau) + \frac{1}{3} R_{ij}(\mathbf{r}, \tau) R_{ij}(\mathbf{r}, \tau). \quad (4)$$

$$H_{ijkl}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^4} \int d\tau d^3\mathbf{r} e^{i\omega\tau} e^{-i\mathbf{k}\cdot\mathbf{r}} R_{ijkl}(\mathbf{r}, \tau). \quad (5)$$

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^4} \int d\tau d^3\mathbf{r} e^{i\omega\tau} e^{-i\mathbf{k}\cdot\mathbf{r}} \left(R_{ii}(\mathbf{r}, \tau) R_{jj}(\mathbf{r}, \tau) + \frac{1}{3} R_{ij}(\mathbf{r}, \tau) R_{ij}(\mathbf{r}, \tau) \right). \quad (6)$$

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^4} \int d\tau d^3\mathbf{r} e^{i\omega\tau} e^{-i\mathbf{k}\cdot\mathbf{r}} \int \frac{d^3q}{(2\pi)^3} \frac{d^3p}{(2\pi)^3} e^{i(\mathbf{q}+\mathbf{p})\cdot\mathbf{r}} \times \left[F_{ii}(\mathbf{q}, \tau) F_{jj}(\mathbf{p}, \tau) + \frac{1}{3} F_{ij}(\mathbf{q}, \tau) F_{ij}(\mathbf{p}, \tau) \right]. \quad (7)$$

The $\mathbf{r}$ integral gives

$$\int d^3r e^{-i\mathbf{k}\cdot\mathbf{r}} e^{i(\mathbf{q}+\mathbf{p})\cdot\mathbf{r}} = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{q} - \mathbf{p}), \quad (8)$$

so integrating over $\mathbf{p}$ and renaming $\mathbf{q} \rightarrow \mathbf{k}_1$, $\mathbf{u} = \mathbf{k} - \mathbf{k}_1$ with $u = |\mathbf{u}|$, we obtain

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int d\tau e^{i\omega\tau} \int d^3k_1 \left[F_{ii}(\mathbf{k}_1, \tau) F_{jj}(\mathbf{u}, \tau) + \frac{1}{3} F_{ij}(\mathbf{k}_1, \tau) F_{ij}(\mathbf{u}, \tau) \right]. \quad (9)$$

Next insert the temporal Fourier representation

$$F_{ab}(\mathbf{k}, \tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{F}_{ab}(\mathbf{k}, \omega), \quad (10)$$

which gives

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int d^3k_1 \int d\tau e^{i\omega\tau} \int \frac{d\omega_1}{2\pi} \frac{d\omega_2}{2\pi} e^{-i(\omega_1+\omega_2)\tau} \times \left[\tilde{F}_{ii}(\mathbf{k}_1, \omega_1) \tilde{F}_{jj}(\mathbf{u}, \omega_2) + \frac{1}{3} \tilde{F}_{ij}(\mathbf{k}_1, \omega_1) \tilde{F}_{ij}(\mathbf{u}, \omega_2) \right]. \quad (11)$$

The τ integral is

$$\int d\tau e^{i\omega\tau} e^{-i(\omega_1+\omega_2)\tau} = 2\pi \delta(\omega - \omega_1 - \omega_2), \quad (12)$$

so integrating over ω_2 with $\omega_2 = \omega - \omega_1$ yields the convolution form

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3k_1 \int d\omega_1 \left[\tilde{F}_{ii}(\mathbf{k}_1, \omega_1) \tilde{F}_{jj}(\mathbf{u}, \omega - \omega_1) + \frac{1}{3} \tilde{F}_{ij}(\mathbf{k}_1, \omega_1) \tilde{F}_{ij}(\mathbf{u}, \omega - \omega_1) \right]. \quad (13)$$

Now substitute the isotropic incompressible model

$$F_{ij}(\mathbf{k}, \tau) = A(k) P_{ij}(\hat{k}) f(\eta_k, \tau), \quad \tilde{F}_{ij}(\mathbf{k}, \omega) = A(k) P_{ij}(\hat{k}) \tilde{f}(\eta_k, \omega), \quad (14)$$

and evaluate the index contractions explicitly. First,

$$P_{ii}(\hat{k}) = \delta_{ii} - \hat{k}_i \hat{k}_i = 3 - 1 = 2. \quad (15)$$

Second,

$$\begin{aligned} P_{ij}(\hat{k}_1) P_{ij}(\hat{u}) &= (\delta_{ij} - \hat{k}_{1i} \hat{k}_{1j})(\delta_{ij} - \hat{u}_i \hat{u}_j) \\ &= \delta_{ij} \delta_{ij} - \delta_{ij} \hat{u}_i \hat{u}_j - \hat{k}_{1i} \hat{k}_{1j} \delta_{ij} + \hat{k}_{1i} \hat{k}_{1j} \hat{u}_i \hat{u}_j \\ &= 3 - 1 - 1 + (\hat{k}_1 \cdot \hat{u})^2 = 1 + (\hat{k}_1 \cdot \hat{u})^2. \end{aligned} \quad (16)$$

Using Eqs. (15)–(16) in Eq. (13) gives

$$\begin{aligned} H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) &= \frac{1}{(2\pi)^8} \int d^3k_1 \int d\omega_1 A(k_1) A(u) \tilde{f}(\eta_{k_1}, \omega_1) \tilde{f}(\eta_u, \omega - \omega_1) \\ &\quad \times \left[4 + \frac{1}{3} (1 + (\hat{k}_1 \cdot \hat{u})^2) \right]. \end{aligned} \quad (17)$$

This is the Fourier transform valid for all k ; we treat $\mathbf{k} > \mathbf{k}_0$ and $\mathbf{k} \leq \mathbf{k}_0$ separately below.

A. $\mathbf{k} > \mathbf{k}_0$

Using $\mathbf{k} = \mathbf{k}_1 + \mathbf{u}$ so that

$$k^2 = k_1^2 + u^2 + 2k_1 u (\hat{\mathbf{k}}_1 \cdot \hat{\mathbf{u}}) \quad \Rightarrow \quad \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{u}} = \frac{k^2 - k_1^2 - u^2}{2k_1 u}, \quad (18)$$

Eq. (17) becomes

$$\begin{aligned} H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) &= \frac{1}{(2\pi)^8} \int d^3 k_1 \int d\omega_1 A(k_1) A(u) \tilde{f}(\eta_{k_1}, \omega_1) \tilde{f}(\eta_u, \omega - \omega_1) \\ &\times \left[4 + \frac{1}{3} \left(1 + \left(\frac{k^2 - k_1^2 - u^2}{2k_1 u} \right)^2 \right) \right]. \end{aligned} \quad (19)$$

Expanding the bracket,

$$\begin{aligned} \left[4 + \frac{1}{3} \left(1 + \left(\frac{k^2 - k_1^2 - u^2}{2k_1 u} \right)^2 \right) \right] &= \left[4 + \frac{1}{3} + \frac{1}{12k_1^2 u^2} (k^4 + k_1^4 + u^4 - 2k_1^2 k^2 - 2k^2 u^2 + 2k_1^2 u^2) \right] \\ &= \left[\frac{27}{6} + \frac{k^4}{12k_1^2 u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \right]. \end{aligned} \quad (20)$$

Combining, we arrive at (cf. Eq. (29) of [1])

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3 k_1 d\omega_1 A(k_1) A(u) \tilde{f}(\eta_{k_1}, \omega_1) \tilde{f}(\eta_u, \omega - \omega_1) \times \mathcal{K}(k, k_1, u), \quad (21)$$

where we define the geometric kernel

$$\mathcal{K}(k, k_1, u) \equiv \left[\frac{27}{6} + \frac{k^4}{12k_1^2 u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \right]. \quad (22)$$

The temporal Fourier transform of $f(\eta_k, \tau) = \exp(-\frac{\pi}{4}\eta_k^2 \tau^2)$ is

$$\tilde{f}(\eta_k, \omega) = \frac{2}{\eta_k} \exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right). \quad (23)$$

Using $A(k) = C_k \epsilon^{2/3} k^{-11/3} / 4\pi$ and $\eta_k = \frac{1}{\sqrt{2\pi}} \epsilon^{1/3} k^{2/3}$ from [1], the combined factor is

$$g(k, \omega) \equiv A(k) \tilde{f}(\eta_k, \omega) = \frac{2E_k}{4\pi k^2 \eta_k} \exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right), \quad (24)$$

where $E_k = C_k \epsilon^{2/3} k^{-5/3}$. Explicitly introducing integration bounds:

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int_{\mathbb{R}^4} d^3 k_1 d\omega_1 g(k_1, \omega_1) g(u, \omega - \omega_1) \mathcal{K}(k, k_1, u). \quad (25)$$

Moving the k_1 integral to spherical coordinates, with $\mu \equiv \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$ the cosine of the angle between $\mathbf{k}_1$ and $\mathbf{k}$:

$$d^3 k_1 = k_1^2 dk_1 d\Omega_1 = k_1^2 dk_1 d\phi d\mu, \quad \mu \equiv \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1 \in [-1, 1], \quad \phi \in [0, 2\pi), \quad (26)$$

$$u \equiv |\mathbf{k} - \mathbf{k}_1| = \sqrt{k^2 + k_1^2 - 2k k_1 \mu}. \quad (27)$$

After integrating over the azimuthal angle ($\int_0^{2\pi} d\phi = 2\pi$):

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int_0^\infty dk_1 k_1^2 \int_{-1}^1 d\mu \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) g(u, \omega - \omega_1) \mathcal{K}(k, k_1, u). \quad (28)$$

Changing the angular variable from μ to u using Eq. (27),

$$d\mu = \frac{u}{k} \frac{du}{k_1}, \quad (29)$$

with bounds $u \in [|k - k_1|, k + k_1]$:

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int_0^\infty dk_1 \frac{k_1}{k} \int_{|k-k_1|}^{k+k_1} du u \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) g(u, \omega - \omega_1) \mathcal{K}(k, k_1, u). \quad (30)$$

Splitting $\mathcal{K}$ into parts with different powers of u to separate the u -integrals:

$$\begin{aligned} H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) &= \frac{1}{(2\pi)^7} \int_0^\infty dk_1 \frac{k_1}{k} \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) \times \left(\frac{27}{6} - \frac{k^2}{6k_1^2} \right) \int_{|k-k_1|}^{k+k_1} du u g(u, \omega - \omega_1) \\ &+ \frac{1}{(2\pi)^7} \int_0^\infty dk_1 \frac{k_1}{k} \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) \times \left(\frac{k^4}{12k_1^2} + \frac{k_1^2}{12} - \frac{k^2}{6} \right) \int_{|k-k_1|}^{k+k_1} du \frac{1}{u} g(u, \omega - \omega_1) \\ &+ \frac{1}{(2\pi)^7} \int_0^\infty dk_1 \frac{k_1}{k} \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) \times \frac{1}{12k_1^2} \int_{|k-k_1|}^{k+k_1} du u^3 g(u, \omega - \omega_1). \end{aligned} \quad (31)$$

This has the same structure as Eq. (31) of [1].

Introducing the dimensionless variables

$$q \equiv \frac{k_1}{k}, \quad s \equiv \frac{u}{k}, \quad \Omega \equiv \frac{\omega}{\epsilon^{1/3} k^{2/3}}, \quad \Omega_1 \equiv \frac{\omega_1}{\epsilon^{1/3} k^{2/3}}, \quad (32)$$

the integration measures transform as

$$dk_1 = k dq, \quad du = k ds, \quad d\omega_1 = \epsilon^{1/3} k^{2/3} d\Omega_1, \quad (33)$$

and the u -integration bounds become

$$u \in [|k - k_1|, k + k_1] \quad \longrightarrow \quad s \in [|1 - q|, 1 + q]. \quad (34)$$

Using the explicit form of $g(k, \omega)$,

$$g(k, \omega) = \frac{1}{\sqrt{2\pi}} C_k \epsilon^{1/3} k^{-13/3} \exp(-2\Omega^2), \quad (35)$$

the product $g(k_1, \omega_1)g(u, \omega - \omega_1)$ factorizes into an overall dimensional prefactor and a purely dimensionless Gaussian,

$$\begin{aligned} g(k_1, \omega_1)g(u, \omega - \omega_1) &= \frac{1}{2\pi} C_k^2 \epsilon^{2/3} k^{-26/3} q^{-13/3} s^{-13/3} \\ &\times \exp(-2\Omega_1^2 q^{-4/3}) \exp(-2(\Omega - \Omega_1)^2 s^{-4/3}). \end{aligned} \quad (36)$$

Collecting all dimensional factors, the correlator takes the scaling form

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{C_k^2 \epsilon}{256\pi^8} k^{-5} \mathfrak{H}(\Omega), \quad (37)$$

where the entire dependence on physical scales has been absorbed into the dimensionless frequency Ω , and $\mathfrak{H}$ is given by a fully dimensionless integral.

For the split form of the kernel, this dimensionless function reads

$$\begin{aligned} \mathfrak{H}(\Omega) &= \int_0^\infty dq q \int_{-\infty}^\infty d\Omega_1 q^{-13/3} \exp(-2\Omega_1^2 q^{-4/3}) \\ &\times \left[\left(\frac{27}{6} - \frac{1}{6q^2} \right) I_{+1}(q, \Omega, \Omega_1) + \left(\frac{1}{12q^2} + \frac{q^2}{12} - \frac{1}{6} \right) I_{-1}(q, \Omega, \Omega_1) + \frac{1}{12q^2} I_{+3}(q, \Omega, \Omega_1) \right], \end{aligned} \quad (38)$$

with the dimensionless s -integrals

$$I_{+1}(q, \Omega, \Omega_1) \equiv \int_{|1-q|}^{1+q} ds s^{-10/3} \exp\left(-2(\Omega - \Omega_1)^2 s^{-4/3}\right), \quad (39)$$

$$I_{-1}(q, \Omega, \Omega_1) \equiv \int_{|1-q|}^{1+q} ds s^{-16/3} \exp\left(-2(\Omega - \Omega_1)^2 s^{-4/3}\right), \quad (40)$$

$$I_{+3}(q, \Omega, \Omega_1) \equiv \int_{|1-q|}^{1+q} ds s^{-4/3} \exp\left(-2(\Omega - \Omega_1)^2 s^{-4/3}\right). \quad (41)$$

Equivalently, without splitting the kernel,

$$\mathfrak{H}(\Omega) = \int_0^\infty dq q \int_{-\infty}^\infty d\Omega_1 \int_{|1-q|}^{1+q} ds s \Phi(q, \Omega_1; s, \Omega - \Omega_1) \tilde{\mathcal{K}}(q, s), \quad (42)$$

where $\tilde{\mathcal{K}}(q, s)$ is the kernel $\mathcal{K}(k, k_1, u)$ expressed in terms of q and s , and

$$\Phi(q, \Omega_1; s, \Omega - \Omega_1) \equiv q^{-13/3} s^{-13/3} \exp\left(-2\Omega_1^2 q^{-4/3}\right) \exp\left(-2(\Omega - \Omega_1)^2 s^{-4/3}\right). \quad (43)$$

In this form, all integrals are manifestly dimensionless, and the dependence on the turbulent energy injection rate ϵ and the wavenumber k is fully contained in the overall prefactor $C_k^2 \epsilon k^{-5}$.

B. Gogoberidze (2007) Appendix A form

Eq. A.1 of [1]:

$$\int_0^\infty dy \exp(-Ay^2) \exp(-A(y-x)^2) = \left(\frac{\pi}{4(A+B)}\right)^{1/2} \exp\left(\frac{-ABx^2}{A+B}\right) \operatorname{erfc}\left(\frac{-Bx}{\sqrt{A+B}}\right). \quad (44)$$

$$\begin{aligned} H_{ijij}(k, \omega) &= \frac{\varepsilon}{24\pi(2\pi)^{3/2}k} \int_{k_0}^{k_d} dk_1 k_1^{-10/3} \int_{\max(|k_1-k|, k_0)}^{\min(k_1+k, k_d)} du u^{-10/3} \left(k_1^{-4/3} + u^{-4/3}\right)^{-1/2} \\ &\times \left[27 - \frac{k^2}{k_1^2} - \frac{k^2}{u^2} + \frac{k^4}{2k_1^2 u^2} + \frac{k_1^2}{2u^2} + \frac{u^2}{2k_1^2}\right] \\ &\times \exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right) \operatorname{erfc}\left(-\frac{2^{1/2}\varepsilon^{-1/3}\omega}{\left(k_1^{-4/3} + u^{-4/3}\right)^{1/2}}\right). \end{aligned} \quad (45)$$

Here k_0 is the largest scale of the turbulence and k_d is the energy damping scale; the nonlinear integration bounds follow from the finite support of E_k .

Claim: This integral is not divergent as $k \rightarrow 0$.

We define the dimensionless parameters

$$\frac{\varepsilon}{k_0} = M^3, \quad \frac{k_d}{k_0} = R^{3/4}, \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}. \quad (46)$$

$$x = \left(\frac{k_1}{k_0}\right)^{-4/3}, \quad y = \left(\frac{u}{k_0}\right)^{-4/3}, \quad (47)$$

$$k_1 = k_0 x^{-3/4}, \quad u = k_0 y^{-3/4}. \quad (48)$$

$$dk_1 = -\frac{3}{4}k_0 x^{-7/4} dx, \quad du = -\frac{3}{4}k_0 y^{-7/4} dy. \quad (49)$$

$$k_1^{-10/3} = k_0^{-10/3} x^{5/2}, \quad u^{-10/3} = k_0^{-10/3} y^{5/2}, \quad (50)$$

$$dk_1 k_1^{-10/3} = -\frac{3}{4} k_0^{-7/3} x^{3/4} dx, \quad du u^{-10/3} = -\frac{3}{4} k_0^{-7/3} y^{3/4} dy. \quad (51)$$

$$k_1^{-4/3} + u^{-4/3} = k_0^{-4/3} (x + y), \quad \left(k_1^{-4/3} + u^{-4/3} \right)^{-1/2} = k_0^{2/3} (x + y)^{-1/2}. \quad (52)$$

$$k_1 \in [k_0, k_d] \implies x \in [R^{-1}, 1]. \quad (53)$$

$$\tilde{k}_1 = \frac{k_1}{k_0} = x^{-3/4}, \quad \tilde{u} = \frac{u}{k_0} = y^{-3/4}, \quad (54)$$

$$\tilde{u}_{\min} = \max(|\tilde{k}_1 - p|, 1), \quad \tilde{u}_{\max} = \min(\tilde{k}_1 + p, R^{3/4}), \quad (55)$$

$$y_{\min} = \tilde{u}_{\max}^{-4/3}, \quad y_{\max} = \tilde{u}_{\min}^{-4/3}. \quad (56)$$

$$27 - \frac{k^2}{k_1^2} - \frac{k^2}{u^2} + \frac{k^4}{2k_1^2 u^2} + \frac{k_1^2}{2u^2} + \frac{u^2}{2k_1^2} = 27 - p^2 x^{3/2} - p^2 y^{3/2} + \frac{p^4}{2} x^{3/2} y^{3/2} + \frac{1}{2} x^{-3/2} y^{3/2} + \frac{1}{2} y^{-3/2} x^{3/2}. \quad (57)$$

$$\exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right) = \exp\left(-\frac{2xy}{x+y} \frac{q^2}{M^2}\right), \quad (58)$$

$$\operatorname{erfc}\left(-\frac{2^{1/2}\varepsilon^{-1/3}\omega}{(k_1^{-4/3} + u^{-4/3})^{1/2}}\right) = \operatorname{erfc}\left(-\frac{\sqrt{2}q}{M\sqrt{x+y}}\right). \quad (59)$$

$$\begin{aligned} H_{ijij}(p, q) &= \frac{3M^3 k_0^{-4}}{256(2\pi)^{3/2} p} \int_{R^{-1}}^1 dx x^{3/4} \int_{y_{\min}(x)}^{y_{\max}(x)} dy y^{3/4} (x+y)^{-1/2} \\ &\times \left[27 - p^2 x^{3/2} - p^2 y^{3/2} + \frac{p^4}{2} x^{3/2} y^{3/2} + \frac{1}{2} x^{-3/2} y^{3/2} + \frac{1}{2} y^{-3/2} x^{3/2} \right] \\ &\times \exp\left(-\frac{2xy}{x+y} \frac{q^2}{M^2}\right) \operatorname{erfc}\left(-\frac{\sqrt{2}q}{M\sqrt{x+y}}\right). \end{aligned} \quad (60)$$

$$y_{\max} = \min(|x^{-3/4} - p|^{-4/3}, 1), \quad y_{\min} = \max((x^{-3/4} + p)^{-4/3}, R^{-1}). \quad (61)$$

II. MONOCHROMATIC AND WHITE-NOISE SPECTRA

We now consider two alternative models for the turbulence power spectrum and carry out the same GW-kernel derivation in each case. The first model replaces the Kolmogorov spectrum by a monochromatic (delta-function) distribution in wavenumber; the second takes the real-space two-point function to be spatially uncorrelated (delta function in $\mathbf{r}$), which corresponds to a flat (white-noise) spectrum in k -space. For each spatial model we work out both the Kraichnan stationary decorrelation and the decaying-turbulence temporal model, giving four cases in total.

The derivation follows the same sequence of steps as in Sec. 1: (i) write H_{ijij} as a double convolution in both $\mathbf{k}$ and ω , (ii) insert the spectral model, (iii) perform the tensor contractions to get $\mathcal{K}$, (iv) switch to spherical coordinates and the variable $u = |\mathbf{k} - \mathbf{k}_1|$, (v) evaluate or simplify the temporal integral, and (vi) introduce dimensionless variables.

A. Monochromatic energy spectrum $E(k) = E_0 \delta(k - k_0)$

We set

$$E(k) = E_0 \delta(k - k_0), \quad A(k) = \frac{E(k)}{4\pi k^2} = \frac{E_0}{4\pi k^2} \delta(k - k_0), \quad (62)$$

so the spectral tensor is

$$F_{ij}(\mathbf{k}, \tau) = \frac{E_0 \delta(k - k_0)}{4\pi k^2} P_{ij}(\hat{k}) f(\tau), \quad \tilde{F}_{ij}(\mathbf{k}, \omega) = \frac{E_0 \delta(k - k_0)}{4\pi k^2} P_{ij}(\hat{k}) \tilde{f}(\omega), \quad (63)$$

where $f(\tau)$ (and its Fourier transform $\tilde{f}(\omega)$) carries the temporal decorrelation, to be specified below. The total kinetic energy density is $\int_0^\infty E(k) dk = E_0$.

Starting from Eq. (17) and writing $d^3 k_1 = k_1^2 dk_1 d\phi d\mu$ with $\mu \equiv \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$, the radial k_1 integral immediately yields

$$\int_0^\infty k_1^2 dk_1 A(k_1) = \int_0^\infty dk_1 \frac{E_0 \delta(k_1 - k_0)}{4\pi} = \frac{E_0}{4\pi}. \quad (64)$$

With k_1 fixed to k_0 , the second spectral factor evaluates on $u \equiv |\mathbf{k} - \mathbf{k}_1| = \sqrt{k^2 + k_0^2 - 2kk_0\mu}$:

$$A(u) = \frac{E_0 \delta(u - k_0)}{4\pi u^2}. \quad (65)$$

We change the angular variable from μ to u using Eq. (27), i.e. $u du = kk_0 |d\mu|$, and perform the azimuthal integral ($\int_0^{2\pi} d\phi = 2\pi$):

$$2\pi \int_{-1}^1 d\mu \frac{E_0 \delta(u(\mu) - k_0)}{4\pi u(\mu)^2} = \frac{E_0}{2} \int_{|k-k_0|}^{k+k_0} \frac{du}{kk_0} \frac{\delta(u - k_0)}{u} = \frac{E_0}{2kk_0^2} \Theta(2k_0 - k). \quad (66)$$

The Heaviside factor $\Theta(2k_0 - k)$ enforces the triangle inequality: the root $u = k_0$ lies in the range $[|k - k_0|, k + k_0]$ only when $k \leq 2k_0$, i.e. $p \leq 2$ with $p \equiv k/k_0$.

The geometric kernel at $k_1 = u = k_0$ evaluates as (using $p = k/k_0$)

$$\mathcal{K}(k, k_0, k_0) = \frac{27}{6} + \frac{k^4}{12k_0^4} + \frac{k_0^2}{12k_0^2} + \frac{k_0^2}{12k_0^2} - \frac{k^2}{6k_0^2} - \frac{k^2}{6k_0^2} = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12} \equiv \mathcal{K}_0(p). \quad (67)$$

Collecting Eqs. (64)–(67) in Eq. (17),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{E_0^2 \Theta(2k_0 - k)}{2(2\pi)^8 k k_0^2} \mathcal{K}_0(p) \underbrace{\int_{-\infty}^{\infty} d\omega_1 \tilde{f}(\omega_1) \tilde{f}(\omega - \omega_1)}_{I_\omega(\omega)}. \quad (68)$$

All spatial structure has been reduced to the explicit prefactor; the remaining task is to evaluate the temporal convolution I_ω .

1. Kraichnan decorrelation

The Kraichnan model gives $\tilde{f}(\omega) = (2/\eta_0) \exp(-\omega^2/\pi\eta_0^2)$ with the fixed rate $\eta_0 \equiv \eta_{k_0}$. The convolution of two identical Gaussians is itself Gaussian:

$$\begin{aligned} I_\omega(\omega) &= \frac{4}{\eta_0^2} \int_{-\infty}^{\infty} d\omega_1 \exp\left(-\frac{\omega_1^2 + (\omega - \omega_1)^2}{\pi\eta_0^2}\right) \\ &= \frac{4}{\eta_0^2} e^{-\omega^2/(2\pi\eta_0^2)} \int_{-\infty}^{\infty} d\omega_1 \exp\left(-\frac{2(\omega_1 - \omega/2)^2}{\pi\eta_0^2}\right) \\ &= \frac{2\sqrt{2}\pi}{\eta_0} \exp\left(-\frac{\omega^2}{2\pi\eta_0^2}\right). \end{aligned} \quad (69)$$

Defining the dimensionless frequency $\Omega \equiv \omega/\eta_0$ and substituting into Eq. (68),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{\sqrt{2} E_0^2}{(2\pi)^7 k k_0^2 \eta_0} \mathcal{K}_0(p) \Theta(2-p) e^{-\Omega^2/(2\pi)}. \quad (70)$$

Writing $k = pk_0$, the entire result separates into a dimensional prefactor times a dimensionless expression:

$$\boxed{\mathfrak{H}_{\text{Kraichnan}}^{(\delta k)}(p, \Omega) = \frac{\mathcal{K}_0(p)}{p} \Theta(2-p) e^{-\Omega^2/(2\pi)},} \quad (71)$$

with $\mathcal{K}_0(p) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}$ from Eq. (67). There is no remaining integral: both delta functions have been evaluated in closed form.

2. Decay model

For the decaying-turbulence model the temporal kernel is

$$\tilde{f}(\omega) = \hat{g}(\omega\tau_1), \quad \hat{g}(q) = e^{iq}(-iq)^{-5/3} \Gamma(\tfrac{1}{3}, -iq), \quad (72)$$

where τ_1 is the characteristic decay time at scale k_0 . Substituting into Eq. (68) and defining $q = \omega\tau_1$, $q_1 = \omega_1\tau_1$, so that $d\omega_1 = dq_1/\tau_1$:

$$I_\omega(\omega) = \int_{-\infty}^{\infty} d\omega_1 \hat{g}(\omega_1\tau_1) \hat{g}((\omega - \omega_1)\tau_1) = \frac{1}{\tau_1} \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(q - q_1). \quad (73)$$

Because $\hat{g}$ has no closed-form convolution, this one-dimensional integral must be retained. Inserting Eq. (73) into Eq. (68),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{E_0^2 \Theta(2k_0 - k)}{2(2\pi)^8 k k_0^2 \tau_1} \mathcal{K}_0(p) \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(q - q_1). \quad (74)$$

The dimensionless result is

$$\boxed{\mathfrak{H}_{\text{decay}}^{(\delta k)}(p, q) = \frac{\mathcal{K}_0(p)}{p} \Theta(2-p) \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(q - q_1).} \quad (75)$$

The spatial integration is again fully resolved by the two delta functions; only a single one-dimensional frequency integral remains.

B. Real-space delta-function correlations

We now consider a model in which the spatial part of the two-point velocity correlation is a delta function in $\mathbf{r}$:

$$R_{ij}(\mathbf{r}, \tau) = v_0^2 \delta^{(3)}(\mathbf{r}) P_{ij} T(\tau), \quad (76)$$

where v_0^2 is the velocity variance and $T(\tau)$ is the (to-be-specified) temporal decorrelation function. Taking the spatial Fourier transform,

$$F_{ij}(\mathbf{k}, \tau) = v_0^2 P_{ij}(\hat{k}) T(\tau), \quad \tilde{F}_{ij}(\mathbf{k}, \omega) = v_0^2 P_{ij}(\hat{k}) \tilde{T}(\omega), \quad (77)$$

so the spectral amplitude is k -independent: $A(k) = v_0^2$. This corresponds to an energy spectrum $E(k) \propto k^2$ (blue, ultraviolet-dominated), and in practice the integrals must be cut off between the outer scale k_0 and a dissipation scale k_d .

The tensor contractions proceed identically to Sec. 1 and yield the same kernel $\mathcal{K}(k, k_1, u)$. With $A(k_1) = A(u) = v_0^2$ and the temporal factor $\tilde{T}(\omega)$ independent of k , Eq. (9) becomes

$$H_{ijij}(\mathbf{k}, \omega) = \frac{v_0^4}{(2\pi)^7} \underbrace{\int_{-\infty}^{\infty} d\omega_1 \tilde{T}(\omega_1) \tilde{T}(\omega - \omega_1)}_{J(\omega)} \int_{k_0}^{k_d} \frac{k_1}{k} dk_1 \int_{\max(|k-k_1|, k_0)}^{\min(k+k_1, k_d)} u du \mathcal{K}(k, k_1, u). \quad (78)$$

The temporal convolution $J(\omega)$ and the double spatial integral are fully decoupled because $\tilde{T}$ does not depend on k_1 or u . Introducing dimensionless integration variables

$$z \equiv \frac{k_1}{k_0}, \quad y \equiv \frac{u}{k_0}, \quad p \equiv \frac{k}{k_0}, \quad R^{3/4} \equiv \frac{k_d}{k_0}, \quad (79)$$

so $dk_1 = k_0 dz$, $du = k_0 dy$, the spatial integral becomes

$$k_0^2 \int_1^{R^{3/4}} \frac{z}{p} dz \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y), \quad (80)$$

where $\tilde{\mathcal{K}}(p, z, y)$ is the geometric kernel of Eq. (19) written in units of k_0 :

$$\tilde{\mathcal{K}}(p, z, y) = \left[\frac{27}{6} + \frac{p^4}{12z^2y^2} + \frac{z^2}{12y^2} + \frac{y^2}{12z^2} - \frac{p^2}{6y^2} - \frac{p^2}{6z^2} \right]. \quad (81)$$

1. Kraichnan decorrelation

For a spatially uncorrelated field there is no scale-dependent sweeping rate; we adopt a fixed decorrelation rate η_0 and set $\tilde{T}(\omega) = (2/\eta_0) \exp(-\omega^2/\pi\eta_0^2)$. The temporal convolution is identical in form to Eq. (69):

$$J_{\text{Kraichnan}}(\omega) = \frac{2\sqrt{2}\pi}{\eta_0} \exp\left(-\frac{\omega^2}{2\pi\eta_0^2}\right), \quad (82)$$

which is now a k -independent closed form. Substituting Eqs. (82) and (80) into Eq. (78) and defining $\Omega \equiv \omega/\eta_0$:

$$H_{ijij}(\mathbf{k}, \omega) = \frac{\sqrt{2} v_0^4 k_0^2}{(2\pi)^6 \eta_0} e^{-\Omega^2/(2\pi)} k_0^2 \int_1^{R^{3/4}} \frac{z}{p} dz \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y). \quad (83)$$

The dimensionless integral is

$$\mathfrak{H}_{\text{Kraichnan}}^{(\delta r)}(p, \Omega) = e^{-\Omega^2/(2\pi)} \int_1^{R^{3/4}} \frac{z}{p} dz \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y). \quad (84)$$

The frequency dependence is an explicit Gaussian factor, while the shape in p is determined by the two-dimensional wavenumber integral.

2. Decay model

For the decaying-turbulence model, $\tilde{T}(\omega) = \hat{g}(\omega\tau_1)$ with the same $\hat{g}$ as in Sec. 3.1.2. The temporal convolution is again a one-dimensional integral with no closed form:

$$J_{\text{decay}}(\omega) = \int_{-\infty}^{\infty} d\omega_1 \hat{g}(\omega_1\tau_1) \hat{g}((\omega - \omega_1)\tau_1) = \frac{1}{\tau_1} \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(q - q_1), \quad q \equiv \omega\tau_1. \quad (85)$$

Substituting into Eq. (78),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{v_0^4 k_0^2}{(2\pi)^7 \tau_1} \mathfrak{H}_{\text{decay}}^{(\delta r)}(p, q), \quad (86)$$

where the fully dimensionless expression is now genuinely three-dimensional:

$$\mathfrak{H}_{\text{decay}}^{(\delta r)}(p, q) = \int_1^{R^{3/4}} \frac{z}{p} dz \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(q - q_1). \quad (87)$$

Because J_{decay} is independent of z and y (since $\tilde{T}$ carries no k -dependence), the three integrals are separable: the double wavevector integral and the frequency convolution may be computed independently.

Summary of the four dimensionless integrals

Spectrum	Decorrelation	Dimensionless integral
$E_0\delta(k - k_0)$	Kraichnan	Eq. (71): closed form, no integral
$E_0\delta(k - k_0)$	Decay	Eq. (75): 1D frequency integral
$\delta^{(3)}(\mathbf{r})$	Kraichnan	Eq. (84): 2D wavevector integral
$\delta^{(3)}(\mathbf{r})$	Decay	Eq. (87): 2D wavevector $\times$ 1D frequency

III. NUMERICAL RESULTS: $f_{\text{peak}}^{\text{GW}}(\tau_*, M_0)$

IV. SPECTRUM SURVEY FOR ALL DERIVED MODELS

In this section we plot, on a common convention, the dimensionless GW energy-density spectrum

$$\Omega_{\text{GW}}(p) \propto p^3 H(p, p), \quad p = k_{\text{GW}}/k_0, \quad (88)$$

along the sound-cone diagonal $q = p$ for every model derived in this paper. The p^2 factor relative to the characteristic strain $h_c^2(p) = p H(p, p)$ is the phase-space measure d^3k ; together with causality and a finite white-noise source it predicts $\Omega_{\text{GW}} \sim p^3$ in the IR, followed by a peak near $p \sim 1$ and a decay in the UV. A dotted grey line shows the p^3 reference; cases whose IR slope deviates from 3 by more than ± 0.5 are flagged in the build log of `Notebooks/all_spectra.py`. Models with a $1/p$ pole at the origin (monochromatic $E_0 \delta(k - k_0)$ source) cannot satisfy the universal scaling because the fluid itself carries no IR power.

[1] G. Gogoberidze, T. Kahniashvili, and A. Kosowsky, *Phys. Rev. D* **76**, 083002 (2007), [arXiv:0705.1733 \[astro-ph\]](#).

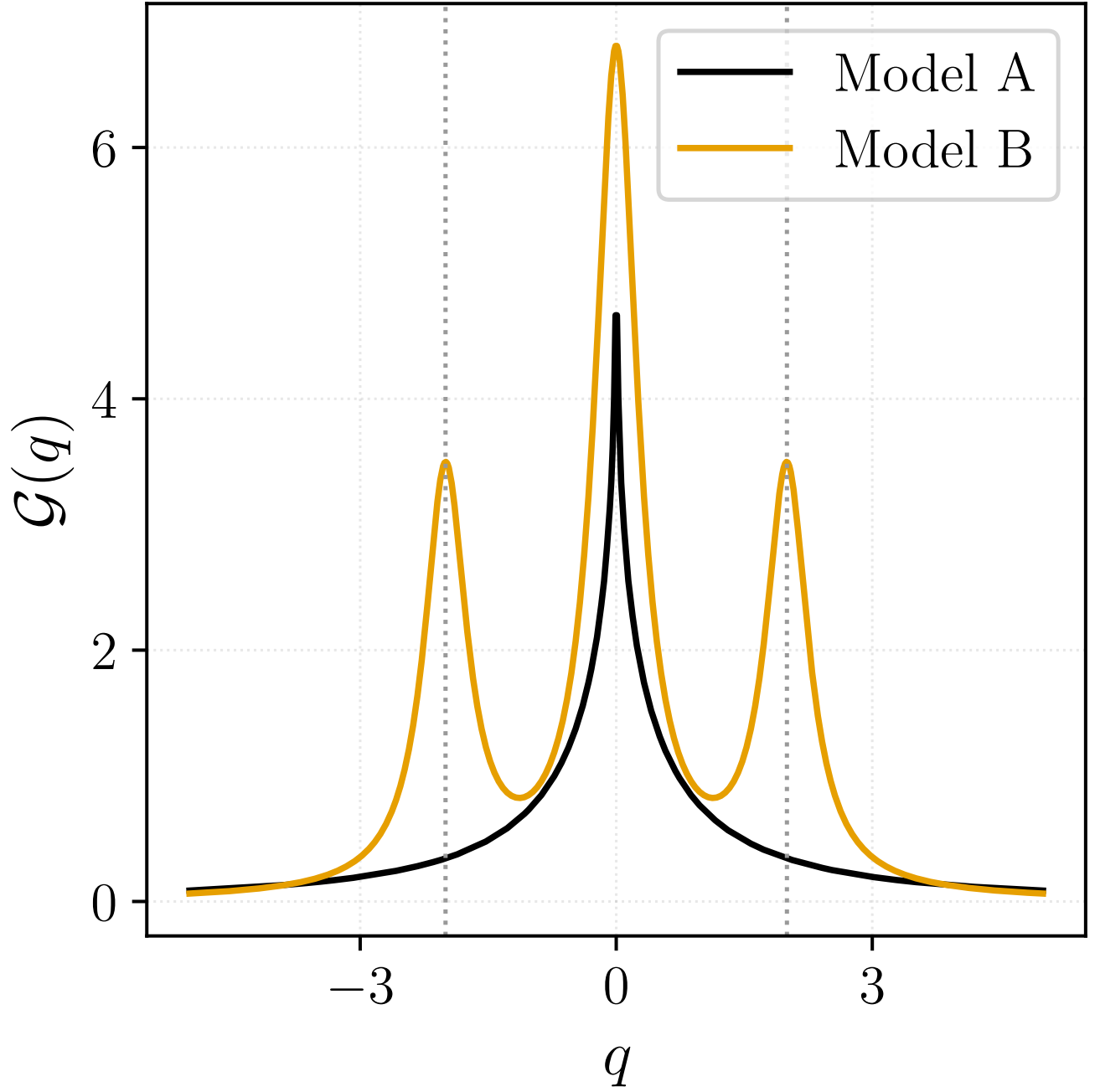


FIG. 1. Dimensionless temporal kernel $\mathcal{G}(q) = \widehat{R^2}(q)$ for the two correlation models considered in Sec. III. Model A uses the random-phase BK2016 form $R_A(\sigma) = (1 + |\sigma|)^{-2/3}$; Model B is the coherent eddy $R_B(\sigma) = \cos(\sigma) e^{-\alpha|\sigma|}$ with $\alpha = 0.15$.

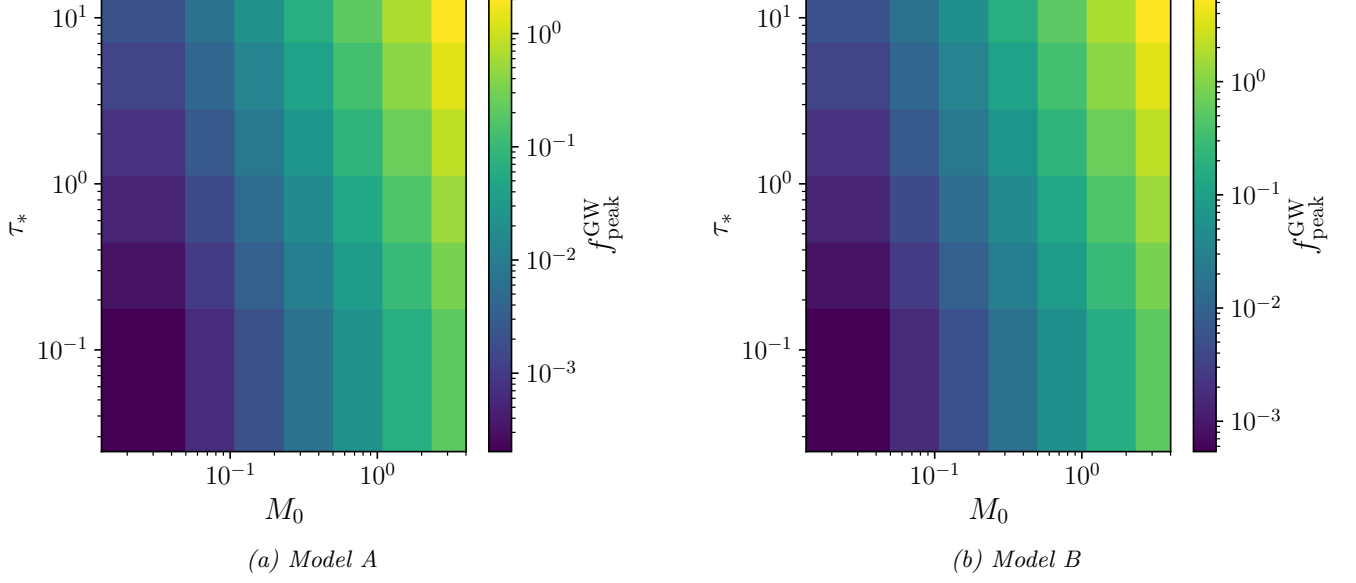


FIG. 2. HD Loitsiansky decay ($q = 2/7$, $p = 10/7$, $\beta = 4$). $f_{\text{peak}}^{\text{GW}}$ on the (τ_*, M_0) plane in units of $k_{p,0}/(2\pi)$. Random-phase BK2016 temporal kernel (a) and coherent eddy temporal kernel (b).

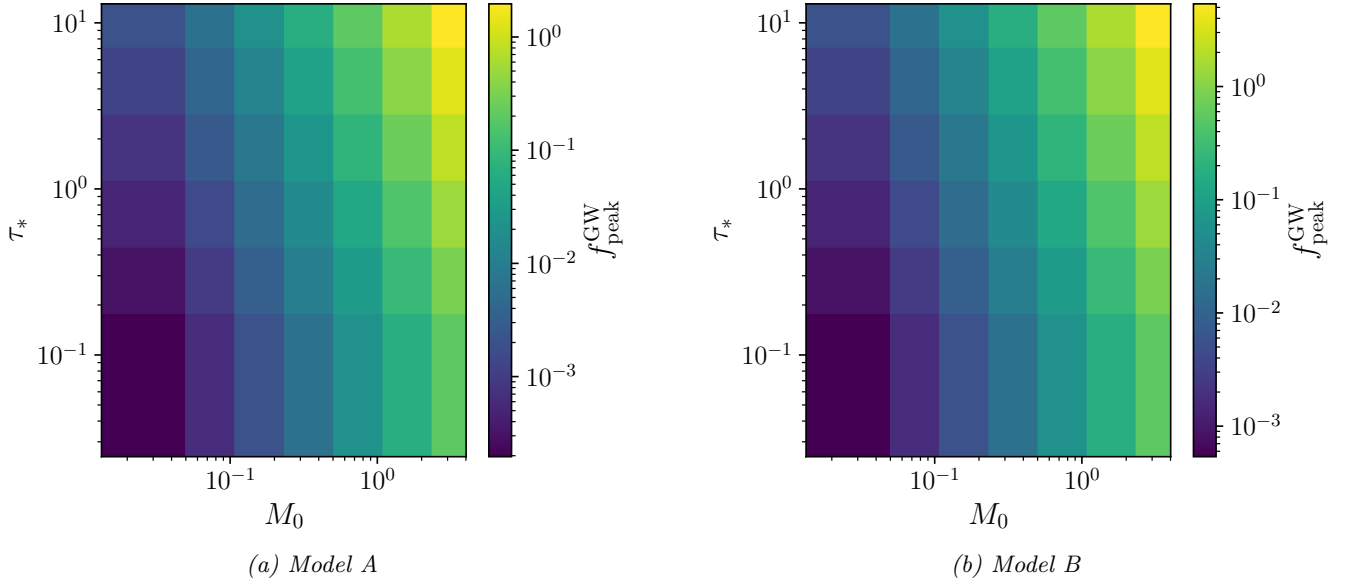


FIG. 3. HD Saffman decay ($q = 1/3$, $p = 4/3$, $\beta = 3$). $f_{\text{peak}}^{\text{GW}}$ on the (τ_*, M_0) plane in units of $k_{p,0}/(2\pi)$. Random-phase BK2016 temporal kernel (a) and coherent eddy temporal kernel (b).

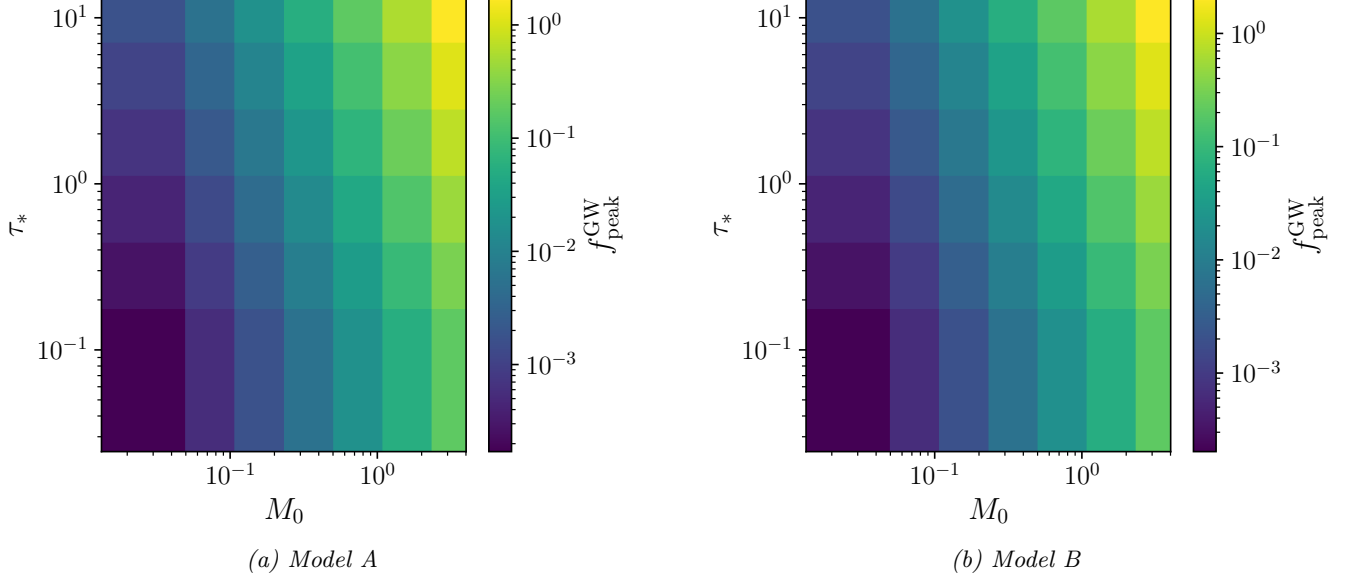


FIG. 4. Nonhelical MHD decay ($q = 1/2$, $p = 1$, $\beta = 1$). $f_{\text{peak}}^{\text{GW}}$ on the (τ_*, M_0) plane in units of $k_{p,0}/(2\pi)$. Random-phase BK2016 temporal kernel (a) and coherent eddy temporal kernel (b).

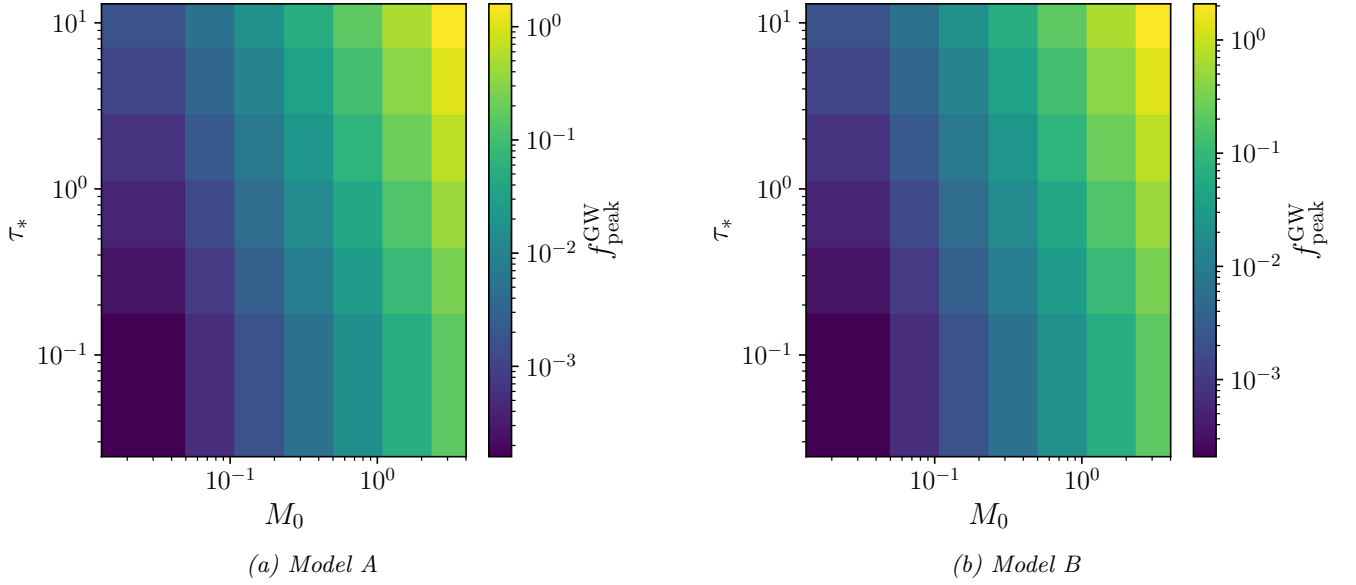


FIG. 5. Helical MHD decay ($q = 2/3$, $p = 2/3$, $\beta = 0$). $f_{\text{peak}}^{\text{GW}}$ on the (τ_*, M_0) plane in units of $k_{p,0}/(2\pi)$. Random-phase BK2016 temporal kernel (a) and coherent eddy temporal kernel (b).

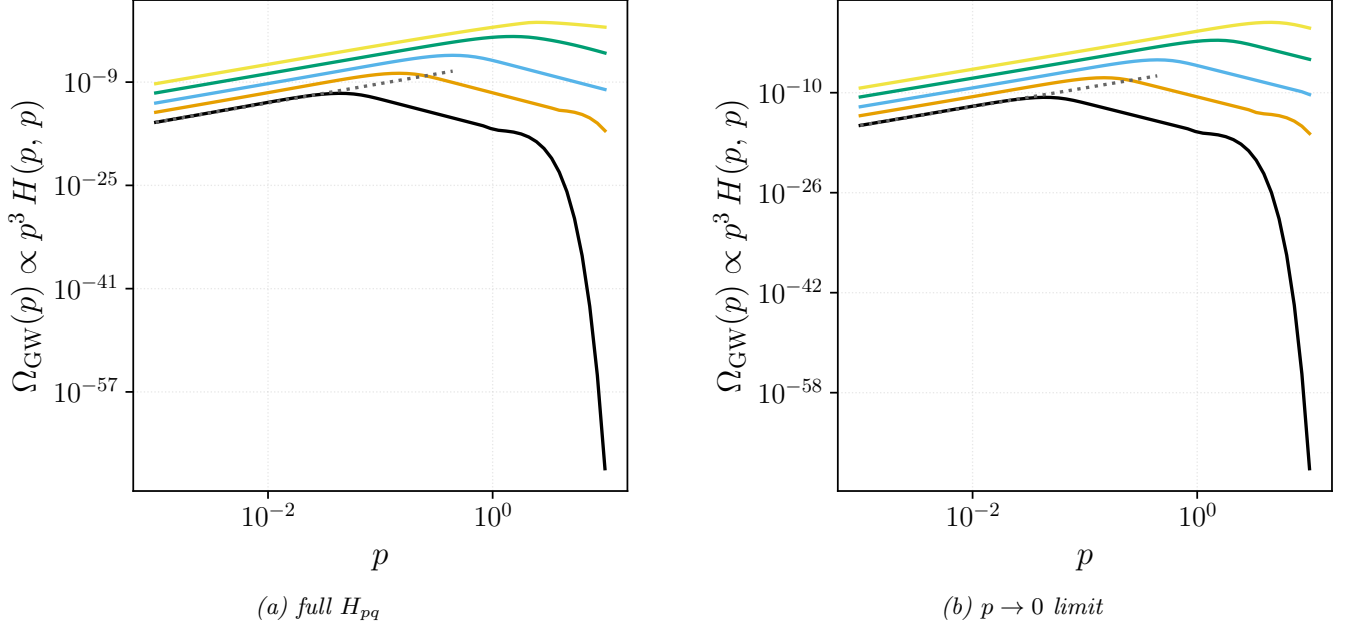


FIG. 6. Kraichnan decorrelation + Kolmogorov spectrum. Five curves per panel correspond to $M = 0.03, 0.1, 0.3, 1.0, 3.0$ at $R = 10^4$. Panel (a) is the full numerical H_{pq} (Sec. 1); panel (b) is the aeroacoustic $p \rightarrow 0$ limit (Kahniashvili et al. 2008). Both give the universal IR slope $\Omega_{\text{GW}} \propto p^3$.

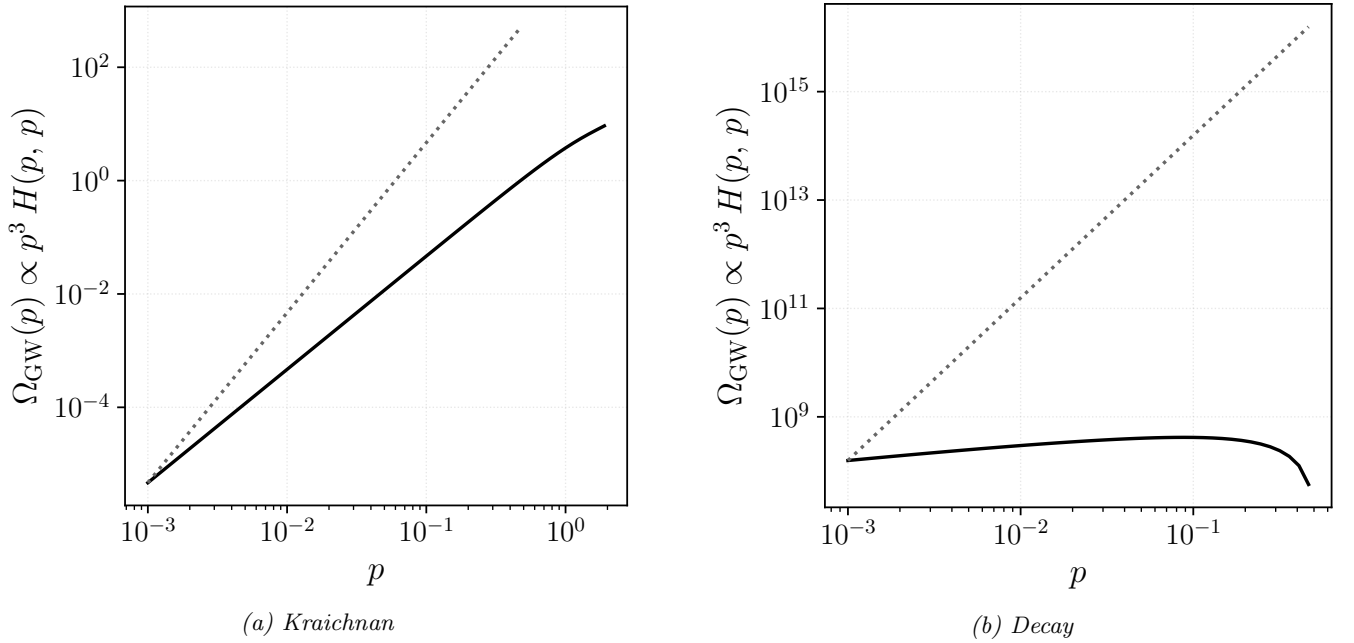


FIG. 7. Monochromatic spectrum $E(k) = E_0 \delta(k - k_0)$ (no free parameters at the dimensionless level). Panel (a): Kraichnan decorrelation, closed form $\Omega_{\text{GW}}(p) \propto p^3 K_0(p)/p \Theta(2-p) e^{-p^2/(2\pi)}$ with IR slope **2** (flagged: $1/p$ pole from the monochromatic source removes one power of p from the universal scaling). Panel (b): decay temporal convolution, IR slope **0.3** (flagged: the temporal-tail extra suppression further reduces the IR power).

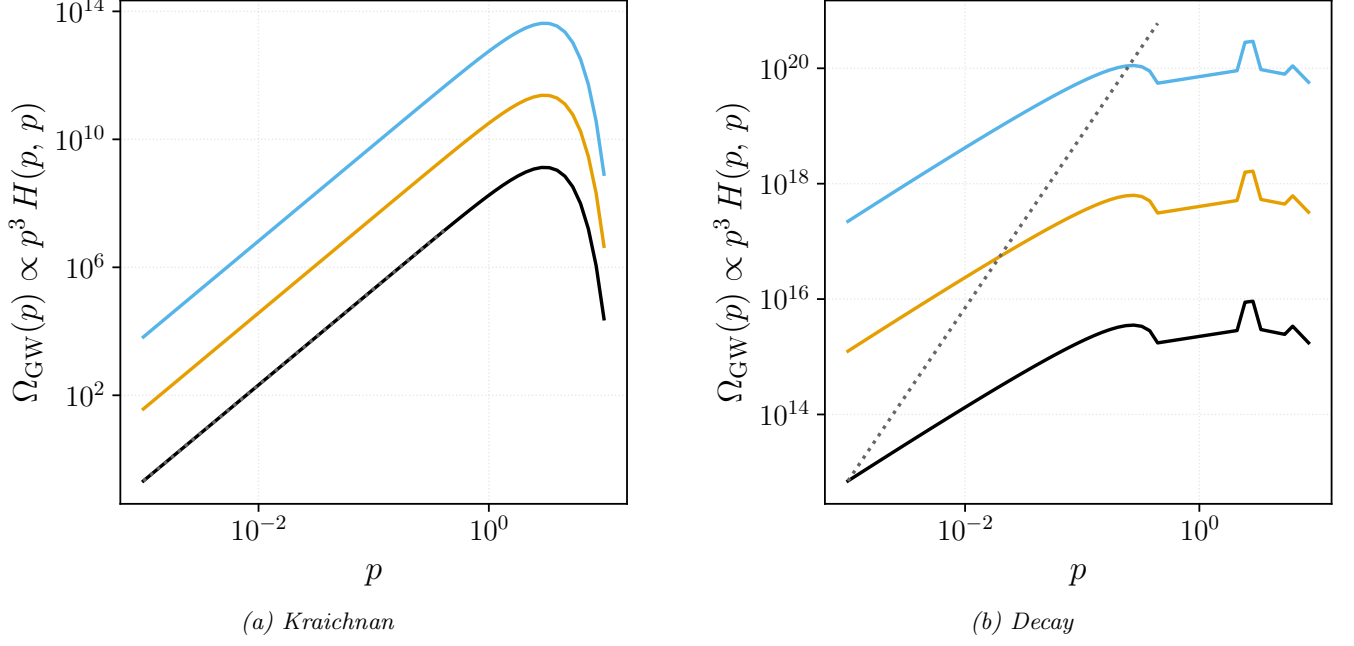


FIG. 8. White-noise real-space correlations $R_{ij} \propto \delta^{(3)}(\mathbf{r})$. Three curves per panel correspond to $R = 10^3, 10^4, 10^5$. Panel (a): Kraichnan decorrelation, matches the universal IR slope **3**. Panel (b): decay temporal model, IR slope **1.3** (flagged: temporal-convolution tails pull power down at small p).